

Integration of Classy-Blocks and OpenFOAM within Blender

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Weekly Progress Report

OpenFOAM GUI

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Introduction

- ◀ **OpenFOAM:** A C++ toolbox for CFD using finite volume methods.
- ◀ **Classy-Blocks:** A Python library for procedural `blockMeshDict` generation, mimicking modern 3D CAD modeling.
- ◀ **Blender Integration:** Utilizing Blender as a frontend for geometry extraction and mesh visualization using `pyvista`.

Research Gap

- ◀ **Gap 1:** Lack of native Blender 5.x support for advanced Python meshing libraries like `classy_blocks`.
- ◀ **Gap 2:** Absence of an integrated “One-Click” solution for geometry extraction, mesh building, and quality reporting within a unified environment.

Problem Statement

- ◀ To develop a Blender Add-on that leverages `classy_blocks` and `pyvista` to automate OpenFOAM case setup, allowing users to extract geometry, generate high-quality meshes, and execute solvers without leaving the Blender interface.

Work Done: STL Extraction & Dynamic Routing

- ◀ **Terrain Export Operator:** Developed `MESH_OT_export_terrain_stl` to extract Blender's active geometric context (e.g. displaced planes) directly into the active OpenFOAM `constant/triSurface` directory.
- ◀ **Dynamic Spec Routing:** Upgraded `build_from_spec` to automatically dispatch geometry build logic. If a block specifies surface mappings, it seamlessly routes generation to the specialized `build_box_block_with_stl` function.

Work Done: Surface Conformity

- ▶ **STL Face Mapping:** Implemented direct integration with `classy_blocks` `project_side()` API. Users can select any local block face (e.g., top, bottom) and logically bind it to the exported TriSurface STL.
- ▶ **End-to-End Conformity:** Fully wired the feature set to ensure that when `blockMeshDict` is generated, the configured block faces automatically conform to the underlying complex terrain representations within OpenFOAM.
- ▶ **Rigorous Safety Tests:** Developed targeted test suites mocking the block builder safely without requiring mesh instantiation, validating side mappings without crashing out-of-bounds `classy_blocks` constraints.